SIM Transformer -- Generalization Across Channel Types (K=1)

Training on a mixed Rician spectrum (K_factor 0-10) with the End-to-End method

1. Objective

The previous K=1 stage validated the pipeline on a single, fixed channel type (K_factor=1). This stage tests and trains for real generalization: can the Transformer controller work well on ANY channel condition, from pure Rayleigh (no line-of-sight) to near-line-of-sight, not just one fixed statistical type -- directly addressing the project's goal of adapting to varying channel conditions.

2. Dataset

20,000 Rician channels generated, each with an independently random K_factor drawn uniformly in [0, 10]. K_factor=0 corresponds to pure Rayleigh fading (no line-of-sight); K_factor=10 is a near-line-of-sight channel. This spans the realistic range of propagation conditions a user could experience. 18,000 samples for training, 2,000 held out for validation.

3. Method

Same End-to-End method (Method 2) validated in the previous stage: the Transformer is trained directly on the physical rate objective (no classical-optimizer labels needed), now exposed to the full spectrum of channel types during training instead of a single fixed type.

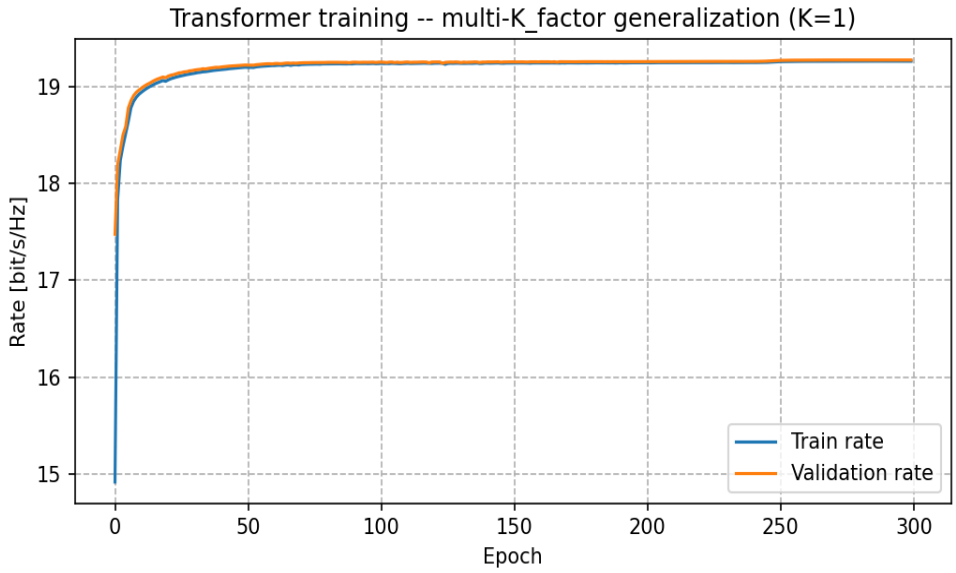


Figure 1. Training converges smoothly; train and validation curves overlap almost perfectly -- no overfitting, stable convergence to ~19.27 bit/s/Hz (continuous, pre-quantization).

4. Results -- Does Performance Hold Across Channel Types?

On 2,000 validation channels (never seen during training, spanning the full K_factor range):

K_factor range	Channel type	Samples	Mean rate (bit/s/Hz)
[0, 2)	Near pure Rayleigh (no LoS)	432	16.50
[2, 4)	Rayleigh-dominant	396	16.39

[4, 6)	Balanced	429	16.39
[6, 8)	LoS-dominant	357	16.39
[8, 10)	Near pure LoS	386	16.38
All	**Full spectrum**	**2000**	**16.41**

The gap between the best and worst K_factor group is only 0.12 bit/s/Hz -- essentially negligible. The model performs nearly identically regardless of the channel type.

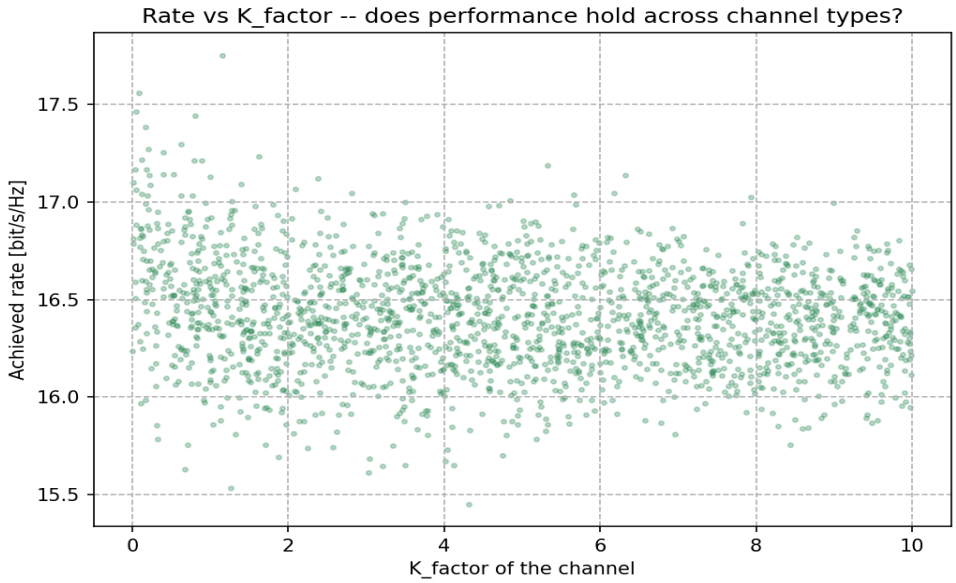


Figure 2. Achieved rate as a function of the channel's K_factor. The scatter stays flat across the whole range -- no degradation at either extreme (pure Rayleigh or near-LoS).

5. Conclusion

Unlike a model trained on a single fixed channel type, this model demonstrates genuine generalization: trained once on a mixed spectrum of channel conditions, it performs consistently well across the entire range, including conditions far from the "average" training case. This directly validates the project's generalization objective for the K=1 stage.

6. Next Steps

- Extend to K>1 (multi-user, with interference), keeping the same multi-K_factor training approach.
- Optionally extend channel diversity further (varying distance/path loss, not just K_factor).
- Integrate the power-amplifier hardware-impairment model (Bussgang).